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MILITARY SPECIFICATION

BORING MACHINES, JIG, VERTICAL, SINGLE COLUMN, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled, single column, vertical spindle jig boring machines which perform machining operations such as boring, drilling, milling, and threading to precision tolerances.

1.2 Classification. The boring machines shall be of the following sizes based on table travel. The size to be furnished shall be as specified (see 6.2.1).

Size	Table Travel (inches)
1118	11 by 18
1630	16 by 30
1828	18 by 28
2036	20 by 36
2236	22 by 36
2848	28 by 48
3660	36 by 60

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the

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issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA ICS 6 - Enclosures for Industrial Controls and Systems.

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 70 - National Electrical Code.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22102-4269.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One Inch Perforated Tape.

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines.

RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

RS-447 - Operational Command and Data Format For Numerically Controlled Machines.

RS-494 - 32-Bit Binary CL Exchange (BCL) Input Format for Numerical Controlled Machines.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006-5009.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current computer numerically controlled (CNC) (three axes simultaneous CNC with manual data input (MDI) capabilities), models capable of boring, drilling, milling, and threading operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified (see 6.2.1), either US or SI units shall be used to graduate measuring and indicating devices. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position. When specified (see 6.2.1), indicating dials shall be graduated to read measurements in both US and SI Units.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and inadvertent damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination of oil, dust, chips, coolants, and airborne particles. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machines order.

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3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize its intended use, performance, or design life. The materials shall have been reclaimed in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Operating controls. Operator's controls for all functions of the machine shall be arranged on a console or pendant, in a location convenient to the operator's normal work station. Controls, such as pushbuttons, dials, and switches, shall be functionally arranged and clearly identified to facilitate use. Emergency controls to stop all machine operations shall be provided. Computer numerical controls shall also meet the requirements given in 3.4.10.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.8 that are applicable to the machine itself. In the event of a conflict between the requirements of ANSI B11.8 and OSHA 29 CFR 1910, the requirements of OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be accessible.

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3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger

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than 0.0001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.3.8 Gears. All gears, shafts, and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The machine shall consist of not less than the base, column, spindle head, spindle and quill assembly, saddle, table, guideways, coolant system, the electrical system, and the CNC system.

3.4.1 Base. The base shall be a casting or steel weldment, ribbed and braced to minimize distortion and vibration. The base shall form a foundation for the entire machine, provide a positive means for securing the column to the base, and have provisions for leveling the machine. The base shall maintain component alignment for assuring the accuracies in table II. When applicable, an access door(s) or removable plate(s) shall be provided for servicing components housed within the base. Precision finished base guideways shall support and guide a saddle. Hydraulic or coolant reservoirs shall be supplied as external units.

3.4.2 Column. The column shall be a casting or steel weldment of box-type construction, internally ribbed and braced to minimize distortion, deflection, and vibration. The column shall accommodate the machining head and have guideways to facilitate vertical movement of the head.

3.4.3 Spindle head. The spindle head shall contain the spindle and rotational drive assembly, quill, head elevating mechanism, and counterbalancing mechanism. The head shall hold the spindle in axial alignment perpendicular to the work holding surface. All spindle head functions shall be numerically controlled. The spindle shall be precision fitted in the head and mounted in anti-friction roller bearings to maintain accuracy. The head shall be held securely in the selected position after a programmed move by a clamping mechanism or servo lock system, interlocked electrically with the drive systems to prevent applying power to a locked or clamped unit.

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3.4.4 Spindle and quill assembly. The spindle shall rotate and be supported in the quill on precision, preloaded, anti-friction bearings. Spindle rotation shall be reversible for tapping operations. The spindle bar shall be alloyed steel, hardened to at least Rockwell C-55, and precision finished to not greater than 18 microinch. Unless otherwise specified (see 6.2.1), the spindle nose shall be the manufacturer's standard quick change type. When specified (see 6.2.1), collets or spindle tool adapters shall be provided for holding the size and type tapered and straight shank tools as specified by the procuring activity. The spindle assembly shall have a power feed system with provisions for selection of normal spindle functions by data input to the control manually or by pre-programming input media.

3.4.5 Saddle and table. The saddle and table shall be fabricated from an iron or iron alloy casting designed to withstand maximum capacity loads. The movement shall be on horizontal guideways for positioning and machining operations. The surface shall be hard or machine finished for a work mounting surface. The worktable shall be equipped with protective covers for the positioning screw and guideways. Automatic means shall be provided to prevent worktable movement after positioning. Overtravel protection devices shall be provided at extremes of worktable travel. A coolant trough shall be integral with the worktable with provisions for draining the coolant to the reservoir.

3.4.6 Position feedback device. The position feedback device, including all related components, hardware, and electrical equipment for the specified axes, shall be provided. The feedback device shall be of the independent type. The feedback device for each axis shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The device shall provide axis travel feedback over the entire range of axis travel of each specified axis. The position feedback device, display console and machine member shall form a closed loop system. The feedback shall be either a linear scale, tape, wire, inductosyn or rack and pinion system.

3.4.7 Drive mechanisms. Means shall be provided to position each moving member in all axes of motion. Manual positioning shall be accomplished by utilizing handwheels and precision leadscrews or ballscrews; movement shall be initiated and controlled by the operator. Power and numerically controlled movement shall be accomplished by solid-state servo drives, electric motors, and precision leadscrews or precision, preloaded, antifriction recirculating ballscrews and nut assemblies. Numerically controlled machines shall also be equipped with position and velocity feedback devices forming a closed loop servo system. Leadscrews or ballscrews shall be hardened and precision ground to provided alignment and positioning accuracy, and be of a size, class, and precision suitable for the application. Leadscrews or ballscrews and nuts shall not exert excessive loads or cause premature deterioration of the feed drive motors or components. Feed rates and positions are to be monitored on numerically controlled machines by a feedback system and transmitted to the control. All slides and screws shall be shielded from contaminants by covers

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or guards. Each slide shall be equipped with automatic power clamping or servo lock, interlocked with the drive controls to prevent the application of power to a clamped or locked mechanism. Unless otherwise specified (see 6.2.1), T-slots shall conform to ANSI B5.1.

3.4.8 Guideways. The guideways on which the saddle, table, and spindle head slide for support and guidance shall be proportioned and constructed to maintain mutual component alignment under heavy cutting loads to minimize wear and friction. Machine ways shall be hardened and ground. Friction-reducing materials, devices, and designs shall be the manufacturer's standard and be of proven performance, suitable for maintaining precision performance of the machine to the specified tolerances. Gibs or other means shall be provided to compensate for wear, as required. All guideways and sliding surfaces shall be protected from chips and other foreign matter.

3.4.9 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to allow full flow of coolant. The system shall have means for draining, cleaning, and shall include a baffle or strainer which is easily removable for cleaning or replacement. Means shall be provided for directional and volumetric control of coolant flow. Means shall be provided for indicating fluid level. The coolant system shall be numerically controlled with a selection of coolant type (mist or flood) and on/off of coolant flow.

3.4.10 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt alternating-current (AC), (with an allowable line fluctuation of $\pm$10 percent), 3-phase, 60-Hz circuit. Initial input voltage shall be as specified (see 6.2.1). Each machine shall have a fused safety disconnect switch or properly rated circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connection to the size conductor dictated by the applicable requirements in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1 and Section 430-7 of ANSI/NFPA-70. Each motor shall have an identification plate to identify manufacturer, model, serial number, voltage, amperage, horsepower, phase, and frequency.

3.4.10.2 Solid-state components. Solid-state design shall be used throughout for electronic components.

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3.4.10.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts.

3.4.10.4 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, and an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described shall not be required.

3.4.11 Computer numerical control (CNC). The machine shall have a microprocessor-type, solid-state, soft-wired, integrated-circuit type, computer numerical control (CNC). The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, two axes (X and Y) positioning and feedrate control, spindle operation, rotary table and other part program directed functions for machines specified herein. Controlled axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of linear axes motion as specified (see 6.2.1). Control features shall include a buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory, manually input data (MDI), or directly from the input media specified in 3.5. The control shall initiate a halt or an error signal should a fault condition occur in the machine or control. All necessary executive program routines shall be furnished in the form of 1-inch, eight-channel binary coded decimal punched tape or other input form appropriate for the input media specified for controls having volatile memory. A line voltage of $\pm$10 percent from normal shall not adversely affect the numerical control system function. The control shall function in ambient temperature ranging from 50 to 120 degrees F and humidity ranging from 5 percent to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safety operating temperature. Control resolution shall be 0.0001 inch for each linear axis and not more than 0.001 degree for each rotary axis. When binary cutter location (BCL) data is specified (see 6.2.1), the numerical control shall have the necessary hardware and software to perform all geometry dependent functions which are ordinarily performed by the post processor.

3.4.11.1 Machine control unit (MCU). The MCU shall provide full control of the machine's X and Y axes. When specified (see 6.2.1), the MCU shall also control a rotary table. The MCU shall control all operations necessary for accomplishing boring, milling, drilling, and tapping to the tolerances specified herein. The control unit enclosure shall comply with ANSI/NEMA ICS 6 provisions.

3.4.11.2 Auxiliary functions. The MCU shall accept command data for operating all functions provided. The auxiliary functions shall be coded in accordance with EIA Standard RS-274.

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3.4.11.3 Feed and speed selection. Spindle speeds, in RPM, and milling feeds, in IPM and mmpm, when applicable, shall be selected by tape command and by MDI.

3.4.11.4 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.11.5 Set zero. The control unit shall have a "set zero" feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.11.6 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.11.7 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 percent and 120 percent of programmed speed. The spindle speed override control shall have a position that allows return of speed to that programmed.

3.4.11.8 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.11.9 Programmed and optional stops. The control unit shall recognize programmed stopping points in the machining cycle. All controlled motions shall be stopped when the code for a programmed stop is read. The optional stop code shall be ignored unless the control is previously set to recognize the command.

3.4.11.10 Coolant control. The coolant system on-off functions shall be controlled during a machining cycle by tape command and by manual control.

3.4.11.11 Programmed-up stop. A programmed-up stop, to use for flat work, shall position the table after the tool clears the work without returning the spindle to full-retract position.

3.4.11.12 Mirror image. The machine shall produce either right- or left-hand from the same tape data through the inversion of X and Y axes. Inversion of each axis shall be by keyboard or switch entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.4.11.13 Acceleration and deceleration. The acceleration and deceleration function shall be accomplished automatically and shall not require programming into the input data.

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3.4.11.14 Dwell. Programmable dwell for the spindle feed shall be provided with a minimum dwell time of 5 seconds.

3.4.11.15 Canned cycles. Automatic and complete machining cycles (canned cycles) shall be provided to permit programming a machining process in a single block using a G-code function for each different cycle. The canned cycles provided shall be for frequently used cycles which shall include various drilling, boring, and milling, cycles. Functions included shall be for rapid advancing the tool to the feed starting point, feed to depth, stop spindle, spindle reversal, rapid withdrawal, and feed out as appropriate for the various types of operations to be preformed by each canned cycle code.

3.4.11.16 Data display. Each machine shall be provided with an optical data display capable of displaying at least 250 characters simultaneously. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, show the following: block display, actual and desired position of each axis (as applicable), feedrate and override conditions, and control and system error messages. For each machine data displays shall indicate a sign (positive or negative) and a minimum of 7 digits.

3.4.11.17 Metric programming. When specified (see 6.2.1), the control unit shall be capable of accepting program data prepared in the U.S. inch system and the International System of Units (SI) for metrics. SI metric units shall be in accordance with FED-STD-376.

3.4.11.18 Postprocessor. When specified (see 6.2.1), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language (see 6.2.1) to the form and format required to operate the machine tool and control. The postpocessor shall be written for use on the type computer specified (see 6.2.1). Documentation sufficient to implement and utilize the postprocessor shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the machine.

3.4.11.19 Axis jog. The control panel shall have manual controls for accomplishing both single and continuous, low- and high-speed jog movements of the slides for each controlled axis in both the plus and minus directions.

3.4.11.20 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, MDI, sequence number search, and optional stop.

3.4.11.21 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches or 999.999mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001mm.

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3.4.11.22 Absolute/incremental input. The control shall have absolute/incremental input capability. The mode shall be selected by a preparatory G-code which may be switched during the tape program. (Not required if binary cutter location (BCL) is specified (see 6.2.1)

3.4.11.23 Part program storage. Unless otherwise specified (see 6.2.1), part program storage shall be in either solid-state or bubble-type memory. The MCU shall be capable of storing in memory the content of, unless otherwise specified (see 6.2.1), not less than 250 feet of post-processed part program tape data. (Not applicable if BCL data input is specified (see 6.2.1)

3.4.11.24 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory, to hold the memory provided at full load for a period of not less than 70 hours when the primary power source fails.

3.4.11.25 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.11.26 Preparatory function (G-codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in EIA Standard RS-494 shall be applied.

3.4.11.27 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in EIA Standard RS-494 shall be applied.

3.4.11.28 Block delete. The control shall have a block delete feature to permit bypassing of blocks of programmed data.

3.4.11.29 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.11.30 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.5 Data input.

3.5.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-227, RS-274, RS-

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358, and RS-447 or when specified (see 6.2.1), RS-244 shall apply in lieu of RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and tape-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust-free area of the numerical control cabinet and shall be easily accessible for cleaning and handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA Standard RS-244.
b. Even parity per EIA Standard RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.5.2 Manual data input. A manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.6 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.7 Alignment and positioning accuracies. The machine alignment and positioning accuracies shall be within the tolerances of table II when checked in accordance with 4.8.3.

3.8 Performance. The machine shall be capable of drilling, boring, and counter boring operations when tested as specified in 4.8.4. The center-to-center dimensions between the holes shall be checked and shall be within ±0.00025 inch. Hole diameters shall be alike within 0.0003 inch total indicating reading (TIR). Concentricity of bored and counter bored diameters shall be within 0.0002 inch TIR.

3.8.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.9 Hydraulic system. The hydraulic system shall meet JIC H-1 and be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure

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delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.10 Standard equipment. All standard equipment required to properly operate the machine shall be furnished.

3.11 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.12 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.13 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.13.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.13.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.14 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the

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nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.15 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.16 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

3.17 Training. Training required by the procuring activity shall be provided as specified in the contract.

3.18 Warranty. Warranty requirements shall be as specified in the contract by the procuring activity. The warranty period shall commence upon completion of the installation and final acceptance test of the machine.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.5).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

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b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under normal operating conditions.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.7, the test in 4.8.1, and the inspection in 4.3. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Acceptance test. Acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. Failure of the machine to pass one or more tests or any part of a test will be cause for rejection.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification, including compliance with all referenced standards.

4.8 Tests. The machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier. The measuring instruments shall have evidence of calibration traceable to the National Bureau of Standards. The numerical control test tapes and computer printouts used shall become the property of the Government.

4.8.1 Machine and control checkout. The manufacturer's standard method for checking proper operation of the machine and control unit and for assisting in trouble shooting the system shall be provided. The check method may consist of built-in devices which are designed to monitor mechanical and electronic functions during operation or system checkout and indicate a malfunction or error, programmed check tape(s) designed to systematically check machine and control operation and indicate a malfunction or error, or a combination of built-in devices and check tape(s) to provide a complete check of the system.

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4.8.2 Operational test. The machine shall be operated at no load for not less than 4 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine that it complies with the alignment tolerance requirements of 3.7

4.8.4 Performance tests. The following test shall be performed to demonstrate the machine's capability to accurately position, bore, and relocate. Under numerical control, drill, bore and counter bore three holes, 3/4 inch diameter by 1 inch deep, in a right angle pattern in a metal plate not less than 1-1/2 inch thick. Counter bored holes shall be 1 inch in diameter by 1/8 inch deep. The plate shall be of a material chosen by the supplier and both faces may be surface ground prior to the test. Predetermined dimensions between the bored holes shall be spaced to utilize as much of the normal working range of the machine as practical, allowing for setup equipment. Test instruments may be of the builder's choice; however, their accuracies shall be traceable to the Bureau of Standards.

4.8.5 Alignment and positioning test. The machine shall be checked in accordance with the machine tool builder's standard test procedure to determine that the machine conforms to the accuracies of table II.

4.8.5.1 Positioning accuracy and repeatability. The positioning accuracy and repeatability for each axis of motion shall be within the tolerances of table II when evaluated in accordance with the NMTBA Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools, Sections 1 and 4, Linear Definition. At least ten random target points, spaced over the full length of the axis, shall be used for evaluating the accuracies. If the system uses a bi-directional programmed approach, each target point shall be evaluated from both directions. Should the axis length exceed the range of the instrument used, the zero position may be shifted to permit checking the full range of travel.

4.8.6 Tapping test. Four holes shall be pilot drilled and taped at programmed locations, each with finished threads to accept 8-32 and 3/4-10 screws. Finished holes shall accept standard 8-32 or 3/4-10 screws freely and threads shall meet class 3 standard for internal threads as set forth in Handbook H28.

4.8.7 Boring test. A minimum of 10 holes shall be pilot drilled, semi-finish bored, and finish bored in a selected pattern covering an area not less than 14 inches in diameter. Four of the holes shall be spaced on 14-inch centers. Different tools shall be used for semi-finish and finish boring operations. Each address shall be to the fourth decimal place with digits other than zero. Finished hole size may range from 1.125 to 2.750 inch.

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4.8.8 Straight-line milling test. The straight line milling test shall be a programmed square or rectangle, not less than 6 inches in length per side with sides parallel to the travel along X and Y axes. Machine angle shall be not less than 2 degrees. The resulting cuts shall be checked for dimensional accuracy, squareness, parallelism, flatness, angularity, and corner locations.

4.8.9 Test program verification. Unless otherwise specified (see 6.2.1), while using one or more test setups devised by the supplier, the machining test program (tape) required under postprocessor (see 3.4.11.18), shall be run on the machining center to verify its ability to receive and perform all program functions without malfunction or error. Test piece(s) shall be either steel, cast iron, or aluminum, while the tooling shall be selected by the supplier as the most suitable for the particular operation performed.

4.8.10 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The numerically controlled jig boring machines covered by this specification are intended for use in any toolroom or production shop where boring, drilling, milling, and tapping operations are performed on jigs, dies, or similar parts to precision tolerances.

6.2 Ordering Data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

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d. If the machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

i. Spindle nose type, if different; collets or tool adapters for straight shank and taper tools, if required (see 3.4.4).

j. T-slot requirement, if different (see 3.4.7).

k. Electrical system, if different; input power, as specified (see 3.4.10).

l. Motor enclosure, if different (see 3.4.10.1).

m. Linear and circular interpolation, as specified see 3.4.11).

n. Binary cutter location (BCL) data, if required (see 3.4.11).

o. Machine control unit (MCU) of rotary table, when specified (see 3.4.11.1).

p. Data display characteristics, if different (see 3.4.11.16).

q. Metric programming, when specified (see 3.4.11.17).

r. Postprocessor (specify computer name and model, form and format), (see 3.4.11.18).

s. Absolute/incremental input, if required (see 3.4.11.22).

t. Solid-state or bubble-type memory, if different (see 3.4.11.23).

u. Memory storage, 250 feet program tape, if different (see 3.4.11.23).

v. One-inch punched tape date input, if not required; EIA Standard RS-244 in lieu of RS-358, if required (see 3.5.1).

w. Tape reader spools, if different, specify (see 3.5.1).

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x. Size and capacity, if different (see 3.6).

y. Additional performance features, if required (see 3.8.1).

z. Standard equipment, if different (see 3.10).

aa. Specify and fully describe optional equipment, if required (see 3.11).

bb. Specify applicable repair parts, if required (see 3.12).

cc. Lubrication chart or plate, if different (see 3.13.1).

dd. Nameplate, if different (see 3.13.2).

ee. Hour meter, if required (see 3.14).

ff. Inspection conditions, if different (see 4.3).

gg. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (see 4.5).

ii. If preservation, packing and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended all inspections be performed within the limits of the continental United States.

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6.6 Subject term (keyword) listing.

Boring machine
Computer numerical control (CNC)
Jig borer
Machine control unit (MCU)
Punched tape
Single column
Vertical spindle

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TABLE I. Sizes and capacities.

Characteristic	Requirements						
	Size 1118	Size 1630	Size 1828	Size 2036	Size 2236	Size 2848	Size 3660
Capacity							
Table working surface, inches, width x length	11x24	17x30	18x32	18x42	22x44	22x54	36x72
Table travel, inches, width x length	11x18	16x30	18x28,13x30 or 18x36	20x36	22x36 or 22x42	28x48	36x60
Table to spindle nose, range, inches	4 to 20	6 to 23	5 to 28	4 to 28-1/2	6 to 27	6 to 27	12 to 37
Head travel on column, inches	12	12	15-1/2	15-1/2	14	14	17
Spindle quill travel, inches	5	5	9	9	9	9	10
Spindle quill diameter, inches	3-1/4	3-3/8	4-7/8	4-5/8	4-5/8	4-7/8	5-1/2
Spindle center to column face, inches	10	12	16	16-1/2	18-1/2	18-1/2	18-1/2
Spindle center to column neck, inches	13-3/4	21	17-1/2	21	24-1/2	30-1/2	44
Speeds and Feeds							
Spindle speeds, number Spindle speed range, RPM	infinite 60 to 2250 or 60 to 3000	infinite 30 to 1500	16 43 to 2000	infinite 43 to 2000	infinite 10 to 2000 60 to 3000	16 30 to 1800	16 40 to 1834
Spindle feeds, number, and range, IPR or IPM	3 - 0.00075, 0.0015, 0.003 or Infinite 0.0009 to 0.036	3 - 0.001, 0.003, 0.006	8 - 0.0005 to 0.010	8 - 0.005 to 0.015 or 0.01 to 30	8 - 0.0005 to 0.015 or 0.01 to 30	8 - 0.0005 to 0.015 or 0.001 to 30	8 - 0.0005 to 0.015 or 0.01 to 30
Head rapid traverse, IPM	40	40	60	60	60	60	60
Worktable and saddle rapid traverse, IPM	60	50	100	150	150	150	60
Worktable and saddle mill rate, range, IPM	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15
Motors, Horsepower							
Spindle drive	2 speed 2/1 or 2	1	3	3	3	3	3

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performanced is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment and positioning accuracies.

Requirements	Permissible Tolerance
Concentricity of spindle nose internal bore, 3/8 inch from nose	0.00005 inch TIR
Concentricity of spindle bore (runout) - proving bar in spindle nose bore:	
1/4 inch from nose	0.0001 inch TIR
15 inches from nose	0.0005 inch TIR
Worktable trueness of travel:	
Size - 1118	0.0002 inch overall
Size - 1630 through 1836	0.0003 inch overall
Size - 2036 through 2242	0.0004 inch overall
Size - 2848 and 3660	0.0005 inch overall
Spindle axis square with worktable surface (tramming method) - largest practicable diameter of tramming circle	0.0003 inch/12-inch radius
Quill motion square to worktable surface (head locked before each reading):	
Quill movement alone	0.00015 inch/10 inches
Head movement alone	0.00015 inch/10 inches
Head and quill combined movement	0.00025 inch overall
Front of worktable parallel with longitudinal motion	0.00015 inch/Ft.
T-slots parallel to each other	0.001 inch/Ft.
Longitudinal motion square to transverse motion	0.0002 inch/Ft. 0.0003 inch overall
Worktable flatness (flat to concave only):	
Crosswise	0.0001 inch/Ft.
Over full width	0.0002 inch/Ft.
Lengthwise	0.0001 inch/Ft.
Over full length	0.0002 inch
Worktable positioning accuracy:	
Transverse	±0.0001 inch/Ft. ±0.0001 inch overall
Longitudinal	±0.0001 inch/Ft. ±0.0002 inch overall
Depth axis positioning accuracy (Z)	±0.0001 inch
Worktable positioning repeatability	
Transverse	±0.00005 inch
Longitudinal	±0.00005 inch
Depth axis repeatability (Z)	±0.0005 inch

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - AR
Navy - MC, AS
Air Force - None

Preparing activity:
DLA-IP

(Project 3411-0070)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-80210B	Boring Machines, Jig Vertical, Single Col , Computer Numerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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